

IMPLEMENTATION PLAN

Project Name

WorkForce360

Document Purpose

This document defines the complete implementation roadmap for WorkForce360.

The goal is to provide a step-by-step development sequence that minimizes rework, reduces AI-generated errors, and ensures each module is built, tested, and integrated before moving to the next phase.

This plan is optimized for AI-assisted development using Antigravity.

DEVELOPMENT PRINCIPLES

Development Approach

Vertical Slice Development

Build one module completely before moving to the next.

Each module must include:

- Database Schema
- Backend APIs
- Frontend UI
- Validation
- Testing
- Integration

before proceeding.

AI Development Strategy

Generate and implement modules individually.

Do NOT generate the entire application in a single prompt.

Recommended Flow:

```
Authentication
↓
Employee Management
↓
Attendance
↓
Payroll
↓
Payslips
↓
Clients
↓
Contracts
↓
Invoices
↓
Expenses
↓
Reports
↓
Settings
```

PHASE 0 - REQUIREMENT VALIDATION

Objective

Ensure all project documents are available and finalized before development begins.

Required Documents

- PRD
 - TRD
 - App Flow Document
 - UI/UX Brief
 - Backend Schema Document
 - Implementation Plan
-

Validation Checklist

Confirm:

- Business Rules Finalized
 - Payroll Logic Finalized
 - Invoice Logic Finalized
 - User Roles Finalized
 - Database Schema Finalized
 - UI Design Finalized
-

PHASE 1 - ENVIRONMENT SETUP

Objective

Prepare all software, services, accounts, and credentials required for development.

Local Software Checklist

Install:

Development Tools

- Node.js (LTS Version)
 - npm
 - Git
 - VS Code
-

Database Tools

- PostgreSQL (Optional Local Installation)
 - Prisma CLI
-

API Testing

- Postman
-

Browser

- Google Chrome
-

Cloud Accounts Checklist

Create Accounts:

Deployment

- Vercel
 - Render
-

Database

- Supabase
-

File Storage

- Cloudinary
-

Source Control

- GitHub
-

Required Credentials

Collect:

Cloudinary

- Cloud Name
 - API Key
 - API Secret
-

Supabase

- Database URL
- Database Password

Vercel

- Deployment Access
-

Render

- Deployment Access
-

Environment Variables Checklist

Prepare:

```
DATABASE_URL=
```

```
JWT_SECRET=
```

```
CLOUDINARY_CLOUD_NAME=
```

```
CLOUDINARY_API_KEY=
```

```
CLOUDINARY_API_SECRET=
```

```
NODE_ENV=
```

PHASE 2 - PROJECT INITIALIZATION

Objective

Create project structure.

Frontend Setup

Initialize:

- Next.js
 - Tailwind CSS
 - Shadcn/UI
 - Axios
 - Recharts
-

Backend Setup

Initialize:

- Node.js
 - Express.js
 - Prisma ORM
-

Repository Setup

Create:

```
workforce360/  
|  
├─ frontend/  
├─ backend/  
└─ docs/
```

Git Setup

Create:

- Main Branch
 - Development Branch
-

PHASE 3 - DATABASE SETUP

Objective

Create database foundation.

Tasks

Create Prisma Schema

Create PostgreSQL Database

Run Initial Migration

Tables To Build

- Users
 - Sessions
 - Employees
 - Designations
 - Family Members
 - Education
 - Employment History
 - Documents
 - Bank Details
 - Salary Structures
 - Salary Revisions
 - Clients
 - Contracts
 - Attendance
 - Payroll
 - Payslips
 - Invoices
 - Expenses
 - Settings
 - Audit Logs
-

Seed Data

Create:

Admin User

Sample Worker

Sample Client

Sample Contract

Sample Expense Category

PHASE 4 - AUTHENTICATION MODULE

Objective

Implement secure authentication.

Backend

Build:

- Login API
 - Password Change API
 - Password Reset API
-

Frontend

Build:

- Login Page
 - Change Password Page
-

Security

Implement:

- JWT Authentication
 - bcrypt Password Hashing
 - Route Protection
-

Testing

Verify:

- Admin Login
 - Worker Login
 - Invalid Credentials
 - Token Expiry
-

PHASE 5 - EMPLOYEE MANAGEMENT MODULE

Objective

Manage employee records.

Backend

Build:

- Employee CRUD APIs
 - Designation APIs
 - Family APIs
 - Education APIs
 - Employment APIs
 - Bank APIs
-

Frontend

Build:

- Employee List
 - Employee Profile
 - Employee Onboarding Wizard
-

Testing

Verify:

- Employee Creation
 - Employee Updates
 - Employee Status Changes
-

PHASE 6 - DOCUMENT MANAGEMENT MODULE

Objective

Manage employee documents.

Backend

Integrate Cloudinary

Build Upload APIs

Frontend

Build:

- Upload Components
 - Preview Components
-

Testing

Verify:

- Upload
 - Replace Document
 - File Validation
-

PHASE 7 - ATTENDANCE MODULE

Objective

Track attendance and overtime.

Backend

Build:

- Attendance APIs
 - Attendance Reports APIs
-

Frontend

Build:

- Mark Attendance Screen
 - Attendance History
-

Testing

Verify:

- Present
 - Half Day
 - Absent
 - OT Hours
-

PHASE 8 - PAYROLL MODULE

Objective

Automate salary calculation.

Backend

Implement:

- Salary Calculation Engine
 - PF Logic
 - ESIC Logic
 - PT Logic
 - Employer Contribution Logic
-

Frontend

Build:

- Payroll Generation
- Payroll Preview
- Payroll History

Testing

Verify:

- Payroll Accuracy
 - Freeze Logic
 - Salary Revision Logic
-

PHASE 9 - PAYSLIP MODULE

Objective

Generate PDF payslips.

Backend

Integrate Puppeteer

Generate PDF Templates

Frontend

Build:

- Payslip Viewer
 - Payslip Download
-

Testing

Verify:

- PDF Generation
 - Download Functionality
-

PHASE 10 - CLIENT MANAGEMENT MODULE

Objective

Manage client records.

Backend

Client CRUD APIs

Frontend

Client List

Client Details

Client Form

Testing

Verify Client Operations

PHASE 11 - CONTRACT MANAGEMENT MODULE

Objective

Manage client contracts.

Backend

Contract CRUD APIs

Frontend

Contract Management Screens

Testing

Verify:

- Contract Creation
 - Contract Status Updates
-

PHASE 12 - INVOICE MODULE

Objective

Generate client invoices.

Backend

Invoice Calculation Engine

GST Calculations

Service Charge Calculations

Frontend

Invoice Generation

Invoice Preview

Invoice History

Testing

Verify:

- Invoice Accuracy
 - Freeze Logic
 - PDF Generation
-

PHASE 13 - EXPENSE MANAGEMENT MODULE

Objective

Track company expenses.

Backend

Expense APIs

Category APIs

Frontend

Expense Screens

Expense Forms

Testing

Verify:

- Expense Creation
 - Expense Reporting
-

PHASE 14 - REPORTS MODULE

Objective

Generate business insights.

Backend

Build:

- Attendance Reports
- Payroll Reports

- Invoice Reports
 - Expense Reports
 - Revenue Reports
 - Profit Reports
-

Frontend

Reports Center

Filters

Exports

Charts

Testing

Verify:

- Report Accuracy
 - Filter Accuracy
-

PHASE 15 - SETTINGS MODULE

Objective

Centralized configuration.

Build

Company Settings

Payroll Settings

User Settings

Testing

Verify Settings Persistence

PHASE 16 - SYSTEM INTEGRATION TESTING

Objective

Validate complete application flow.

Test Scenarios

Employee Onboarding

Attendance Entry

Payroll Generation

Payslip Generation

Client Creation

Contract Creation

Invoice Generation

Expense Recording

Report Generation

PHASE 17 - USER ACCEPTANCE TESTING

Objective

Validate system against business requirements.

Verify

Payroll Matches Existing Manual Calculations

Invoice Matches Existing Invoice Format

Reports Match Business Expectations

PHASE 18 - DEPLOYMENT

Frontend

Deploy to Vercel

Backend

Deploy to Render

Database

Deploy to Supabase PostgreSQL

Storage

Configure Cloudinary

Production Environment Variables

Configure All Secrets

PHASE 19 - POST DEPLOYMENT TESTING

Verify

Authentication

Attendance

Payroll

Payslips

Invoices

Reports

Exports

Cloudinary Uploads

Database Connectivity

MVP COMPLETION CHECKLIST

Authentication ✓

Employee Management ✓

Document Management ✓

Attendance ✓

Payroll ✓

Payslips ✓

Clients ✓

Contracts ✓

Invoices ✓

Expenses ✓

Reports ✓

Settings ✓

Testing ✓

Deployment ✓

Post-Deployment Validation ✓

PROJECT COMPLETE CONDITION

The MVP is considered complete only when:

- All modules are functional
- All business rules are implemented
- Payroll calculations are accurate
- Invoice calculations are accurate
- Reports generate correctly
- Application is deployed
- Post-deployment testing is successful
- Admin and Worker workflows operate without errors